

Problem 11.2

Find the accumulated value of 1 at the end of n years if the force of interest is

$$\delta_t = \frac{1}{1+t}.$$

Solution

The force of interest satisfies

$$\delta_t = \frac{a'(t)}{a(t)} = \frac{d}{dt} \ln a(t).$$

Therefore,

$$\begin{aligned} \ln a(t) &= \int_0^t \delta_x \, dx \\ &= \int_0^t \frac{1}{1+x} \, dx \\ &= \ln(1+x) \Big|_0^t \\ &= \ln(1+t). \end{aligned}$$

Exponentiating both sides gives

$$a(t) = 1+t.$$

Thus, at the end of n years, the accumulated value of 1 is

$$\boxed{a(n) = 1+n}.$$